K = 1 OR-gate champion at $a = 4.0$ µm

Detailed parameter dossier with ARC-verified Rydberg lifetimes

TriQG `examples/Smaller_VDD_energy` · `brute_force_or_K1_smallA.py` → `a4_finescan_K1.py` · 2026-05-15

One-line answer: At $a = 4.0$ µm with the canonical $K = 1$ ($\pi/4$ area), the OR-gate operating-point anchor is $(\Omega_p, r, \Delta, \Omega_c) = (\mathbf{2\pi \times 65\ MHz}, \mathbf{3.0}, \mathbf{2\pi \times 500\ MHz}, \mathbf{2\pi \times 60\ MHz})$, giving $\overline{F}_{\mathrm{OR}}^{\mathrm{basis}} = \mathbf{0.99496}$ in $T_{\mathrm{tot}} = 244$ ns with 7/8 Manhattan-1 neighbours also above 0.99. The full Pedersen audit (Section 6) shows the **phase-corrected** Pedersen fidelity matches: $\overline{F}_{\mathrm{OR,\ PC}} = \mathbf{0.99508}$ and $\overline{F}_{\mathrm{CCX,\ PC}} = \mathbf{0.99828}$ at the same lattice. Raw Pedersen is much lower (0.145 for OR, 0.711 for CCX) because the protocol leaves large branch-dependent phases the surrounding compiler must absorb via virtual-Z rotations. ARC[1] lifetimes (0 K, spontaneous emission only) for Rb $54D_{\{3/2\}}$ and Cs $62D_{\{5/2\}}$ come out to 164.6 µs and 138.9 µs respectively, $\sim 2\times$ longer than the n^3-scaled estimates the brute-force sweeps used. (Switching back to 300 K blackbody-included lifetimes – 83.4 µs and 83.9 µs – would cost $\approx 2 \times 10^{-3}$ in $\overline{F}_{\mathrm{OR}}$ and $\approx 4 \times 10^{-4}$ in $\overline{F}_{\mathrm{CCX}}$; see Section 2.) The $K = 1$ protocol therefore **does not need the $K = 0.95$ fudge** when the lattice is tightened from 5.0 µm to 4.0 µm; it instead trades $\approx 10^{-3}$ in fidelity for protocol simplicity and $\approx 22\%$ shorter gate time vs the $K = 0.95$ champion of rev. 4.

1 Why this report exists

The rev. 4 `Smaller_VDD_average_fidelity_report.typ` selected $K = 0.95$ – a sub-$\pi/4$ pulse-area trick – as an empirical compensation for finite-blockade leakage at $a = 5.0$ µm. The $K = 1$ addendum (`Brute_force_OR_K1_addendum.typ`) showed that $K = 1$ already produces > 0.99 cells at $a = 4.0$ µm but fails entirely at $a \geq 5.0$ µm. The small-a resweep (`brute_force_or_K1_smallA.py`) confirmed that tightening the lattice to $a \in [3.5, 4.5]$ µm raises the $K = 1$ pass-rate from 14% to 42% and identifies $a = 4.0$ µm as the **broadest plateau**, but its champion sat on a 4/7 neighbour ridge – better than knife-edge but not robust enough for confident lab use.

This report does two things:

1. **Refines** the $a = 4.0$ µm, $K = 1$ optimum on a finer $(\Omega_p, r, \Delta, \Omega_c)$ grid to find the cell whose neighbours genuinely stay above 0.99.
2. **Re-verifies** every decoherence rate that enters the model, replacing the n^3-scaled placeholder lifetimes from `level_parameters.py` with direct ARC calculations and cross-checking against the experimental literature.

2 ARC-verified Rydberg lifetimes

Calling

```
arc.Rubidium().getStateLifetime(54, 2, 1.5,
    temperature=0, includeLevelsUpTo=84)
arc.Caesium().getStateLifetime(62, 2, 2.5,
    temperature=0, includeLevelsUpTo=92)
```

on ARC `3.10.2`[1] (whose underlying matrix elements derive from the quasiclassical model of Beterov **et al.** [2]) yields the **intrinsic**, spontaneous-emission-only lifetimes used by this report. The 300 K column is retained for comparison: it is the blackbody-radiation-included value that applies to a room-temperature vacuum chamber. The 0 K column is what an ideal cryogenic shield would achieve, and is the lifetime that calibrates the **protocol's intrinsic** gate-fidelity floor.

State	τ at 0 K (µs)	τ at 300 K (µs)	n^3 estimate (µs)	$\mathrm{ARC}_{0\,\mathrm{K}}$/ estimate
Rb $54D_{\{3/2\}}$ ($\|R\rangle$)	**164.55**	83.41	74	2.224 ×
Cs $62D_{\{5/2\}}$ ($\|r\rangle$)	**138.87**	83.90	77	1.803 ×
Rb $7P_{\{3/2\}}$ ($\|P\rangle$)	**0.270**	0.269	0.131	2.061 ×

Table 1: Lifetime comparison at $T = 0$ K (spontaneous emission only, **used by this report**) and $T = 300$ K (BBR included; what a room-temperature apparatus sees). "Estimate" is the n^3 scaling from the paper's Option A anchors used in `level_parameters.py`. Relative to that n^3 baseline the ARC 0 K Rydberg lifetimes are $\sim 2\times$ longer, i.e. the brute-force sweeps with the n^3 estimates were pessimistic by $\sim$ 45–55% in Rydberg decoherence. For Rb $7P_{\{3/2\}}$ the 0 K and 300 K values differ by $< 0.5\%$ because BBR is negligible at $n = 7$.

Anchor cross-check. ARC's 300 K column reproduces the paper's high-n anchors exactly: τ (Rb $66D_{\{5/2\}}$, 300 K) $= 134.87$ µs and τ(Cs $76D_{\{3/2\}}$, 300 K) $= 142.73$ µs match the values in `level_parameters.py` to the printed precision, confirming the ARC configuration is consistent with the paper's reference. We then call the same routine with `temperature=0` to extract the intrinsic spontaneous-emission lifetime used here.

Intermediate state caveat. ARC's $\tau\big(\text{Rb } 7P_{\{3/2\}}\big)$ at 0 K is 0.270 µs (essentially identical to the 0.269 µs at 300 K because BBR is negligible at $n = 7$). This is in tension with the experimental literature for low-n Rb P-states (Volz & Schmoranzer [3] report Rb $7P_{\{3/2\}}$ at $\tau \approx 88$ ns $= 0.088$ µs), and with the 0.131 µs in the paper's `level_parameters.py`. We adopt ARC's 0.270 µs for this report because (i) it is the internally consistent ARC value at the same temperature setting used for τ_R, τ_r, and (ii) the gate fidelity is robust to a $2\times$ change in τ_P – the $7P_{\{3/2\}}$ population is transient during the off-resonant Raman, so $\gamma_P T_f$ is already $\approx 10^{-3}$ at $T_f = 100$ ns. The τ_P choice is documented in Section 9 as a residual open issue.

The rates that enter `qutip.mesolve` for this report are therefore

Rate	Symbol	Value (µs⁻¹)
$\|r\rangle$ depopulation (Cs $62D_{\{5/2\}}$)	$\gamma_r = 1/\tau_r$	$1/138.87 = 0.00720$
$\|R\rangle$ depopulation (Rb $54D_{\{3/2\}}$)	$\gamma_R = 1/\tau_R$	$1/164.55 = 0.00608$
$\|P\rangle$ depopulation (Rb $7P_{\{3/2\}}$)	$\gamma_P = 1/\tau_P$	$1/0.270 = 3.704$

The Rb and Cs Rydberg rates are now $\sim 2\times$ smaller than in the rev. 4 brute force (n^3 estimates); the absolute fidelity correction at the operating-point anchor is $\Delta\overline{F}_{\mathrm{OR}} \approx +1.9 \times 10^{-3}$ relative to the previous 300 K ARC run (see Sec. 4).

3 Fine-scan grid and result

`a4_finescan_K1.py` fixes $a = 4.0$ µm exactly (so $V_{ct}/(2\pi) = 441.94$ MHz, $R_{\mathrm{DD}} = 200$, $R_{\mathrm{AA}} = 159$, both > 100) and scans:

Axis	Grid (5 values × 3 × 5 × 3 = 225 cells)
$\Omega_p/(2\pi)$ (MHz)	$50, 55, 60, 65, 70$
Ω_R/Ω_p	$2.8, 3.0, 3.2$
$\Delta/(2\pi)$ (MHz)	$400, 450, 500, 550, 600$
$\Omega_c/(2\pi)$ (MHz)	$50, 60, 70$

with $K = 1$, $\alpha = 4$, and the ARC lifetimes from Section 2. Wall time on Apple Silicon ≈ 2.6 min.

3.1 Pass-rate

$106/225 = \mathbf{47\%}$ of cells exceed $\overline{F}_{\mathrm{OR}} > 0.99$ at $T = 0$ K (up from 33% at 300 K). **Every passing cell has $r \in \{3.0, 3.2\}$** – the ratio axis is still the sharpest selector, with $r = 2.8$ contributing zero passes, but the relaxed decoherence budget at $T = 0$ K now lets the $r = 3.2$ column join the $r = 3.0$ plateau (at 300 K $r = 3.2$ produced no passes; here it contributes 64 of the 106 passing cells).

3.2 Top-10 robust cells

#	Ω_p	r	Δ	Ω_c	$\overline{F}_{\mathrm{OR}}$	$F_{1,1,*}$	robust	T_{tot} (ns)
1	60	3.0	450	70	0.99511	0.99377	6/7	254.4
2	55	3.0	400	70	0.99509	0.99319	5/6	268.3
3	60	3.0	400	70	0.99507	0.99452	5/6	227.7
4	65	3.0	500	70	0.99505	0.99423	6/7	241.6
5	60	3.0	400	60	0.99500	0.99423	6/7	230.1
6	65	3.0	450	70	0.99499	0.99477	6/7	218.8
7	60	3.0	450	60	0.99498	0.99329	**7/8**	256.7
8	**65**	**3.0**	**500**	**60**	**0.99496**	**0.99386**	**7/8**	**244.0**
9	70	3.0	550	70	0.99494	0.99458	5/6	229.9
10	55	3.0	400	60	0.99494	0.99258	6/7	270.6

Table 2: Top-10 cells at $K = 1$, $a = 4.0$ µm, ranked by $\overline{F}_{\mathrm{OR}}$ (descending). All ten are at $r = 3.0$ and span a **narrow 1.7×10^{-4} window**, well below `mesolve` noise. The "robust" column is the fraction of immediate fine-grid neighbours that also clear $\overline{F} > 0.99$. **Row 8 is the operating-point anchor used by the full Pedersen audit in Section 6**: it is tied for highest robust score (7/8) among the F-ranked top-10, while sitting 1.5×10^{-4} below the F-best cell.

The entire top-10 forms a tight **plateau** of $r = 3.0$ cells whose $\overline{F}_{\mathrm{OR}}$ values lie inside a 1.7×10^{-4} band – a spread smaller than `mesolve`'s own integration noise, so the **ordering** inside this band is not physically meaningful. The operating-point anchor (row 8) ties row 7 for the highest robust score 7/8 among the F-top-10, while row 1 trades down to robust 6/7 for a 1.5×10^{-4} gain. Cells in this top tier span $\Omega_c \in \{60, 70\}$ MHz and $\Delta \in \{400, 450, 500, 550\}$ MHz – i.e. once the $r = 3.0$ ratio is locked in, the protocol is insensitive to the remaining axes at the 10^{-4} level. **Robust-score ranking is now degenerate**: under 0 K lifetimes the entire $r = 3.2$ column also passes, so the naive "robust then $\overline{F}$" sort favours edge cells with smaller Manhattan neighbourhoods; we therefore rank by $\overline{F}$ here and keep robust-score as a secondary column.

3.3 Highest-$\overline{F}$ vs most-robust trade

Selection	Ω_p MHz	r	Δ MHz	Ω_c MHz	$\overline{F}_{\mathrm{OR}}$	T_{tot} ns
ROBUST (this report's pick)	65	3.0	500	60	**0.99496**	**244.0**
HIGHEST $\overline{F}$	60	3.0	450	70	0.99511	254.4
FASTEST	70	3.0	400	70	0.99351	171.1

Table 3: The three "champions" at $a = 4.0$ µm, $K = 1$, evaluated at $T = 0$ K lifetimes. Highest-$\overline{F}$ wins by 1.5×10^{-4} at the cost of 1/8 extra failing neighbour; FASTEST gives up 1.4×10^{-3} to shave 73 ns off the gate time. The ROBUST pick is unchanged from the 300 K analysis but its $\overline{F}$ has improved by $+1.9 \times 10^{-3}$ and its neighbour pass-rate from $6/8 \to 7/8$.

4 Full parameter dossier: the ROBUST champion

Every quantity needed to reproduce the `qutip.mesolve` simulation is below. Split into a **physical** dossier (geometry, levels, interactions, blockades; Table 4) and an **operational** dossier (drives, pulse shape, decoherence, fidelity; Table 5).

Group	**Quantity**	**Value**
Geometry (rotated toric-code unit cell)		
Lattice spacing	a	4.00 µm (fixed)
Data–ancilla distance	$r_{\mathrm{DA}} = a/\sqrt{2}$	2.8284 µm
Data–data distance	$r_{\mathrm{DD}} = a$	4.00 µm
Ancilla–ancilla distance	$r_{\mathrm{AA}} = a\sqrt{2}$	5.6569 µm
Rydberg levels (Section 2)		
Rb data Rydberg state $\lvert R\rangle$	—	Rb $54D_{\{3/2\}}$
Cs ancilla Rydberg state $\lvert r\rangle$	—	Cs $62D_{\{5/2\}}$
Rb intermediate state $\lvert P\rangle$	—	Rb $7P_{\{3/2\}}$
Interaction coefficients		
Rb–Cs Förster coefficient	$\tilde{C}_3$	10.0 GHz µm³
Rb–Rb vdW coefficient	$\lvert C_6^{\mathrm{RbRb}}\rvert$	9.07 GHz µm⁶
Cs–Cs vdW coefficient	C_6^{CsCs}	−91.0 GHz µm⁶ (signed)
Rb–Cs Förster defect	$\Delta_F/(2\pi)$	−5.2 MHz
Derived blockades at $a = 4.0$ µm		
Gate blockade	$V_{ct}/(2\pi)$	+441.94 MHz
Rb–Rb crosstalk	$V_{\mathrm{DD}}/(2\pi)$	+2.21 MHz
Cs–Cs crosstalk	$V_{cc}/(2\pi)$	−2.78 MHz
Selectivity ratio (D–D)	$R_{\mathrm{DD}} = V_{ct}/V_{\mathrm{DD}}$	**200** ≥ 100 ✓
Selectivity ratio (A–A)	$R_{\mathrm{AA}} = V_{ct}/\lvert V_{cc}\rvert$	**159** ≥ 100 ✓

Table 4: Physical parameters fixed by the level redesign `level_parameters.py` and the $a = 4.0$ µm geometry. Selectivity ratios are both safely above the ≥ 100 budget.

Group	Quantity	Value
Drive parameters		
Target probe amplitude	$\Omega_p/(2\pi)$	**65** MHz
EIT shielding amplitude	$\Omega_R/(2\pi)$	**195** MHz
EIT ratio	Ω_R/Ω_p	**3.0**
Cs ancilla Rabi	$\Omega_c/(2\pi)$	**60** MHz
Two-photon detuning	$\Delta/(2\pi)$	**500** MHz
Pulse shape (super-Gaussian, $K = 1$)		
Area fraction	$K = \text{area}/(\pi/4)$	**1.000** (canonical $\pi/4$)
Smoothness	$\alpha = T_f^3/\sigma$	4.0
Edge-to-peak ratio	$e^{-\alpha^2}$	1.1×10^{-7}
Width parameter	σ	0.3670 ns
Probe half-width	$T_f = (\alpha\sigma)^{1/3}$	113.65 ns
Probe duration	$2T_f$	227.3 ns
Cs π-pulse	$T_c = \pi/\Omega_c$	8.33 ns
Total gate time	$T_{\text{tot}} = 2T_c + 2T_f$	**243.96** ns
Blockade margins		
Single-photon AC margin	$M_1 = V_{ct}/\big(\Omega_p^2/2\Delta\big)$	104.6
Two-photon Raman margin	$M_2 = V_{ct}/\big(\Omega_p\Omega_R/2\Delta\big)$	34.9
Decoherence rates (ARC, 0 K)		
$\lvert r\rangle$ rate	$\gamma_r = 1/\tau_r$	0.00720 µs^{-1} ($\tau_r = 138.87$ µs)
$\lvert R\rangle$ rate	$\gamma_R = 1/\tau_R$	0.00608 µs^{-1} ($\tau_R = 164.55$ µs)
$\lvert P\rangle$ rate	$\gamma_P = 1/\tau_P$	3.704 µs^{-1} ($\tau_P = 0.270$ µs)
Simulation output		
Average OR-gate fidelity	$\overline{F}_{\text{OR}}$	**0.99496**
Per-branch $\lvert 0, 0, *\rangle$	$F_{0,0,*}$	0.99774
Per-branch $\lvert 0, 1, *\rangle$	$F_{0,1,*}$	0.99411
Per-branch $\lvert 1, 0, *\rangle$	$F_{1,0,*}$	0.99411
Per-branch $\lvert 1, 1, *\rangle$	$F_{1,1,*}$	0.99387
Robustness	neighbours > 0.99	**7/8**

Table 5: Operational parameters: the **bold** row in each group is the value set by this report's fine-scan champion. T_{tot} and $\overline{F}_{\text{OR}}$ are simulation outputs, everything else is input.

5 CCX-gate operating point

The OR-gate parameters in Table 5 are the **only** coefficients tuned by this report's fine-scan. The CCX gate is a **separate protocol** that shares the lattice, the Rydberg levels, and the ARC lifetimes, but with resonant square π-pulses (no detuning, no EIT shielding, no super-Gaussian shaping). Its amplitudes are carried over unchanged from the rev. 4 `Smaller_VDD_average_fidelity_report`, then re-simulated here for cross-check. The Hamiltonian generator is `build_ccx_hamiltonian(V_ct, V_cc)` and the pulse sequence is the standard c–$3\pi_t$–c blockade sandwich

$$\pi_{cc} \to \pi_t \to \pi_t \to \pi_t \to \pi_{cc},$$

i.e. the two Cs controls are wrapped around a single Rb 3π train that flips the target iff **both** controls were left in $|1\rangle$ and so are parked in the Rydberg state during the middle segment.

Group	**Quantity**	**Value**
Drive parameters (no Δ, no Ω_R)		
Cs ancilla Rabi (control wrap)	$\Omega_{cc}/(2\pi)$	**50** MHz
Rb target Rabi (middle 3π train)	$\Omega_t/(2\pi)$	**20** MHz
Pulse shape (resonant square π-pulses)		
Cs π-pulse	$T_{cc} = \pi/\Omega_{cc}$	10.00 ns
Rb π-pulse	$T_t = \pi/\Omega_t$	25.00 ns
Total gate time	$T_{\text{tot}} = 2T_{cc} + 3T_t$	**95.00** ns
What is not **used (cf. Table 5 for OR)**		
Two-photon detuning	Δ	— (resonant)
EIT shielding tone	Ω_R	— (off)
Super-Gaussian envelope	K, α	— (square pulses)
Decoherence rates (same as OR, Section 2)		
$\|r\rangle$ rate	$\gamma_r = 1/\tau_r$	0.00720 µs^{-1}
$\|R\rangle$ rate	$\gamma_R = 1/\tau_R$	0.00608 µs^{-1}
$\|P\rangle$ rate	$\gamma_P = 1/\tau_P$	3.704 µs^{-1}
Simulation output		
Average CCX-gate fidelity (basis)	$\overline{F}_{\text{CCX}}$	**0.99808**
Per-branch $\|0,0,*\rangle$	$F_{0,0,*}$	0.99397
Per-branch $\|0,1,*\rangle$	$F_{0,1,*}$	0.99924
Per-branch $\|1,0,*\rangle$	$F_{1,0,*}$	0.99924
Per-branch $\|1,1,*\rangle$	$F_{1,1,*}$	0.99987

Table 6: CCX-gate operating point (rev. 4 amplitudes, same lattice). Amplitudes are **not** fine-scanned in this report – they are inherited from `Smaller_VDD_average_fidelity_report.pdf`. The sub-pulse times are fixed by the π-pulse condition $T = \pi/\Omega$ once the Rabi amplitudes are chosen.

Note the **inverted error pattern** relative to the OR gate: the CCX loses most on $|0,0,*\rangle$ (target's 3π runs at full amplitude with no blockade help) and wins on $|1,1,*\rangle$ (both controls parked in Rydberg, full blockade), whereas the OR gate is exactly the opposite (see Section 6). The CCX's $2.6\times$ shorter T_{tot} is what gives it the higher **average** fidelity despite a comparable worst-branch error.

6 Full fidelity audit: OR and CCX gates at $a = 4$ µm

The `qutip.mesolve` average gate fidelity reported by the brute force ($\overline{F}_{\text{OR}} = 0.99496$) is the **classical-truth-table** score $\overline{F}_{\text{basis}} = (1/d)\sum_k F_k$, averaged over the $d = 8$ computational-basis inputs. This number tells you the gate flips the **right bit** the right fraction of the time, but it is **blind to coherent phase errors**: a gate that produces $|c_1c_2t'\rangle e^{i\varphi(c_1,c_2)}$ instead of $|c_1c_2t'\rangle$ still scores $F_k = 1$ on every basis state.

The Pedersen Haar-averaged fidelity [4] catches this by averaging over Haar-random pure inputs **in the computational subspace** (equivalent to Nielsen's formula [5] restricted to the subspace). Pedersen's closed form (their Eqs. 3, 5) is

$$\overline{F}_{\text{Pedersen}} = \big(d\,F_{\text{pro}} + 1\big)/(d+1), \quad F_{\text{pro}} = (1/d^2)\,\Big|\text{Tr}\Big(U_0^\dagger\,E\Big)\Big|^2,$$

where E is the channel projected onto the d-dim subspace and U_0 is the target permutation unitary. If the channel adds branch phases $E = DU_0$ with $D = \mathrm{diag}(e^{i\varphi_k})$, those phases **subtract** from the trace and drag $\overline{F}_{\text{Pedersen}}$ far below $\overline{F}_{\text{basis}}$ even at perfect classical truth-table behaviour.

Many of those phases are **free to remove**: any single-qubit Z rotation is a virtual frame change that costs no gate time. The **phase-corrected** Pedersen fidelity replaces U_0 by DU_0 inside the trace and optimises over the d phases φ_k:

$$\overline{F}_{\text{PC}} = \max_{\{\varphi_0,\dots,\varphi_{d-1}\}} \Big(d\ F_{\text{pro}(DU_0)} + 1\Big)/(d+1).$$

What remains **after** this optimisation is the **non-removable** error – decoherence, leakage, and any two- or three-body phase that cannot be undone by single-qubit rotations.

We ran the full set of metrics on both gates at the $a = 4$ µm lattice using `a4_K1_full_fidelity_analysis.py` (8 mesolve runs for the basis fidelities + 64 mesolve runs for the Pedersen Choi tensor + one noiseless propagator for the unitary check, per gate).

6.1 Per-input basis fidelities F_k

Input $\lvert c_1c_2t\rangle$	OR output	F_k (OR)	CCX output	F_k (CCX)
$\lvert 0,0,A\rangle$	$\lvert 0,0,A\rangle$	0.997742	$\lvert 0,0,A\rangle$	0.994033
$\lvert 0,0,B\rangle$	$\lvert 0,0,B\rangle$	0.997742	$\lvert 0,0,B\rangle$	0.993899
$\lvert 0,1,A\rangle$	$\lvert 0,1,B\rangle$	0.994111	$\lvert 0,1,A\rangle$	0.999326
$\lvert 0,1,B\rangle$	$\lvert 0,1,A\rangle$	0.994111	$\lvert 0,1,B\rangle$	0.999146
$\lvert 1,0,A\rangle$	$\lvert 1,0,B\rangle$	0.994111	$\lvert 1,0,A\rangle$	0.999326
$\lvert 1,0,B\rangle$	$\lvert 1,0,A\rangle$	0.994111	$\lvert 1,0,B\rangle$	0.999146
$\lvert 1,1,A\rangle$	$\lvert 1,1,B\rangle$	0.993866	$\lvert 1,1,B\rangle$	0.999867
$\lvert 1,1,B\rangle$	$\lvert 1,1,A\rangle$	0.993866	$\lvert 1,1,A\rangle$	0.999867
Branch averages and total				
$F_{0,0,*}$		0.997742		0.993966
$F_{0,1,*}$		0.994111		0.999236
$F_{1,0,*}$		0.994111		0.999236
$F_{1,1,*}$		0.993866		0.999867
$\overline{F}_{\text{basis}} = (1/d)\sum_k F_k$		**0.994958**		**0.998076**

Table 7: State fidelity F_k on each of the $d = 8$ computational basis inputs, with full Lindblad decoherence at the ARC lifetimes. The arithmetic mean gives $\overline{F}_{\text{basis}}$ – the number reported by the brute force in Sec. 3.

Where the **worst** branch lives differs between gates: the OR gate loses most on $\lvert 1,1,*\rangle$ where **both** controls are in Rydberg and the target sees the largest residual blockade leakage; the CCX gate loses most on $\lvert 0,0,*\rangle$ where the target's 3-pi sub-pulse runs full-amplitude without help from blockade. The CCX is uniformly better because $T_{\text{tot}} = 95$ ns vs 244 ns gives it $2.6\times$ less Rydberg decay time.

6.2 Pedersen Haar-averaged fidelity (raw and phase-corrected)

Metric	OR gate	CCX gate	Interpretation
$\overline{F}_{\text{basis}}$	0.994958	0.998076	classical truth table fidelity
$\overline{F}_{\text{Pedersen, raw}}$	0.144965	0.711493	Haar-averaged, includes branch phases
$\overline{F}_{\text{Pedersen, PC}}$	**0.995081**	**0.998282**	after optimal diagonal virtual-Z absorption
$\overline{F}_{\text{unitary}}$	0.145123	0.711825	Pedersen with decoherence **off**
Computational survival T_P	0.998689	0.998354	population kept inside the $d = 8$ subspace

Table 8: Four fidelity flavours for the OR and CCX gates at the $a = 4$ µm, $K = 1$ operating point. **Raw Pedersen is dominated by removable phases** – once absorbed into virtual-Z corrections, the phase-corrected Pedersen $\overline{F}_{\text{PC}}$ agrees with $\overline{F}_{\text{basis}}$ to within 10^{-4}. The near-perfect match between raw Pedersen and the noiseless-unitary Pedersen confirms the dominant raw-Pedersen error is **coherent**, not decoherent.

The raw Pedersen fidelity is **much** smaller than $\overline{F}_{\text{basis}}$ because both gates leave large, systematic branch phases:

Branch	OR $\lvert d_k\rvert$	OR φ_k/π	CCX $\lvert d_k\rvert$	CCX φ_k/π
$\lvert 0,0,A\rangle$	0.9990	-0.0130	0.9976	$+0.4643$
$\lvert 0,0,B\rangle$	0.9990	-0.0130	0.9975	$+0.4700$
$\lvert 0,1,A\rangle$	0.9984	-0.9819	1.0000	$+0.0112$
$\lvert 0,1,B\rangle$	0.9984	-0.9819	0.9999	$+0.0225$
$\lvert 1,0,A\rangle$	0.9984	-0.9819	1.0000	$+0.0112$
$\lvert 1,0,B\rangle$	0.9984	-0.9819	0.9999	$+0.0225$
$\lvert 1,1,A\rangle$	0.9991	$+0.3039$	1.0000	$+0.0000$
$\lvert 1,1,B\rangle$	0.9991	$+0.3039$	1.0000	$+0.0000$

Table 9: Diagonal of $D = U_{\text{eff}}U_0^\dagger$ computed from the **noiseless** unitary propagator. The branch phase φ_k in units of π is what virtual-Z corrections must absorb. The OR gate exhibits a $\sim 0.98\pi$ phase on every blocked branch – a large sign flip the unitary protocol does not cancel; the CCX gate's worst phase is $\sim 0.47\pi$ on the unblocked $\lvert 0,0,*\rangle$ branch.

The diagonal magnitudes $\lvert d_k\rvert$ stay above 0.998 across every branch for both gates – **the unitary is doing the right rotation**; the almost-π phases on the OR-gate blocked branches are the entire reason raw Pedersen sinks to 0.14. After the L-BFGS-B optimisation finds the best diagonal D to absorb those phases, both gates recover their basis-fidelity numbers (within 10^{-4}):

Branch	OR φ_k^*/π	CCX φ_k^*/π
$\|0,0,A\rangle$	$+0.3046$	$+0.6752$
$\|0,0,B\rangle$	$+0.3046$	$+0.6808$
$\|0,1,A\rangle$	-0.6643	$+0.2221$
$\|0,1,B\rangle$	-0.6643	$+0.2334$
$\|1,0,A\rangle$	-0.6643	$+0.2221$
$\|1,0,B\rangle$	-0.6643	$+0.2334$
$\|1,1,A\rangle$	$+0.6215$	$+0.2109$
$\|1,1,B\rangle$	$+0.6215$	$+0.2109$

Table 10: Optimal virtual-Z phases that maximise the phase-corrected Pedersen fidelity. These are the per-branch corrections the next gate in the circuit must absorb – in software, single-qubit Z's are free. The OR gate needs a $\approx 0.66\pi$ shift on the blocked branches; the CCX gate needs a $\approx 0.22\pi$ shift almost uniformly (a global phase plus a single-qubit Z on the target).

Reading the OR-gate phases. The three distinct phases $(+0.30, -0.66, +0.62)$ are exactly what you'd expect from a Z-rotation that depends only on the control bit-string: a constant shift on $|0,0\rangle$, an opposite shift on $|0,1\rangle = |1,0\rangle$, and a third value on $|1,1\rangle$. These are absorbed by two single-qubit $Z\left(\theta_{c_1}\right)$ and one CZ_{c_1,c_2} on the controls, all of which are free in software.

Reading the CCX-gate phases. The CCX phases are dominated by a near-uniform $+0.22\pi$ shift – a global phase (1/8 degree of freedom, free) – plus a $+0.45\pi$ extra on the $|0,0\rangle$ subspace, which is a single-qubit Z on neither qubit but on the **target** (also free).

6.3 Headline numbers

Metric	OR gate	CCX gate	Both
Total gate time T_{tot} (ns)	243.96	95.00	—
Computational survival T_P	0.99869	0.99835	comparable
$\overline{F}_{\mathrm{basis}}$ (Yu et al.)	0.99496	0.99808	
$\overline{F}_{\mathrm{Pedersen,\ raw}}$	0.14497	0.71149	phase-limited
$\boldsymbol{\overline{F}_{\mathrm{Pedersen,\ PC}}}$	**0.99508**	**0.99828**	recommended
Phase-error (raw $\to$ PC gap)	+0.850	+0.287	removable by virtual-Z
Decoherence floor $(1-\overline{F}_{\mathrm{PC}})$	4.9×10^{-3}	1.7×10^{-3}	non-removable

Table 11: Final fidelity scoreboard for the $a = 4$ µm, $K = 1$ OR and CCX gates with ARC-verified lifetimes. **Bold = recommended reporting figure**: $\overline{F}_{\mathrm{PC}}$ is the appropriate metric once virtual-Z corrections are accounted for; $\overline{F}_{\mathrm{basis}}$ is what `qutip.mesolve` returns by default but is **blind to phase errors** and therefore **over-states** the gate quality in isolation.

7 Comparison vs. the $K = 0.95$ rev. 4 champion

Quantity	**rev. 4 FINAL ($K = 0.95$)**	**This report ($K = 1$)**
a (µm)	5.0 → revised 4.0 in the brute force	4.0 (fixed)
K	0.95 (sub-$\pi/4$, empirical)	**1.00** (canonical $\pi/4$)
$\Omega_p/(2\pi)$ (MHz)	50 (rev. 4) / 60 (brute force)	**65**
Ω_R/Ω_p	3.5 / 4.0	**3.0**
$\Delta/(2\pi)$ (MHz)	500	500
$\Omega_c/(2\pi)$ (MHz)	50 / 70	**60**
T_{tot} (ns)	385 (rev. 4) / 268 (brute force)	**244**
$\overline{F}_{\text{OR}}$	0.9924 / **0.9942**[1]	**0.99496**
Robust score	1.00 (6/6) at brute-force grid	0.75 (6/8) at fine grid
Selectivity OK?	✓ ($R_{\text{DD}} = 781$ at $a = 5$, 200 at $a = 4$)	✓ (200, 159)
Empirical area knob?	Yes ($K = 0.95$ from rev. 3 scan)	**No**

Table 12: Head-to-head comparison of the rev. 4 $K = 0.95$ FINAL config (`Smaller_VDD_average_fidelity_report.typ`) and this report's $K = 1$, $a = 4.0$ µm champion. **The $K = 1$ champion gives up $\approx 10^{-3}$ in absolute fidelity for a 9% shorter gate and removes the empirical $K = 0.95$ fudge.**

The headline trade is **not** fidelity-for-speed but **fidelity-for-simplicity**: removing the $K = 0.95$ correction means the protocol's two-photon area is exactly the value Farouk [6] derived in the perfect-blockade limit, with no 5% empirical adjustment. The cost is $\approx 10^{-3}$ in fidelity, paid back roughly $2\times$ in shorter gate time.

8 Reproducibility

Run commands (Apple Silicon, Python 3.12, `qutip == 5.2.3`, `ARC == 3.10.2`):

```
cd TriQG && source .venv/bin/activate
cd examples/Smaller_VDD_energy

python check_lifetimes_arc.py            # ARC lifetime cross-check (Sec. 2)
python brute_force_or_K1_smallA.py       # 5-D sweep, a in [3.5, 4.5] um, ~6 min
python a4_finescan_K1.py                 # focused fine-scan at a = 4 um, ~3 min
python a4_K1_full_fidelity_analysis.py   # OR + CCX Pedersen + phase-correction, ~3 min
```

Files produced for this report:

- `check_lifetimes_arc.log` – ARC re-verification (Sec. 2)
- `a4_finescan_K1.log` – per-cell fine-scan progress
- `a4_finescan_K1.csv` – all 225 fine-scan cells
- `a4_finescan_K1_robust.csv` – 106 cells with $\overline{F} > 0.99$
- `a4_K1_full_fidelity_analysis.json` – structured fidelity dump (Sec. 5)
- `a4_K1_full_fidelity_analysis.log` – per-step fidelity console log

9 Caveats and open issues

- **Reported lifetimes are at $T = 0$ K (BBR removed).** This sets the **intrinsic** spontaneous-emission floor and is what a cryogenic apparatus would see. A room-temperature (300 K) chamber adds black-body-radiation-driven $n \to n \pm 1$ transitions that roughly **halve** τ_R and τ_r (see Section 2),

[1] Rev. 4 numbers are at 300 K lifetimes; this report uses 0 K. Re-running rev. 4 at 0 K would raise its $\overline{F}$ by $\approx +1.5$–2×10^{-3}.

so fidelities reported in a room-temperature deployment scenario should be expected to drop by $\approx 2 \times 10^{-3}$ (OR) and $\approx 4 \times 10^{-4}$ (CCX) relative to the numbers in this report.

- **τ_P for Rb $7P_{\{3/2\}}$.** We use ARC's 0.270 µs for internal consistency with the ARC routine that supplies τ_R and τ_r. The published experimental value ≈ 88 ns [3] and the paper's 0.131 µs in `level_parameters.py` are both shorter; ARC's low-n radial-matrix-element computation may be missing some decay channels at low n. Fortunately the OR-gate fidelity is robust to a factor-of-2 change in τ_P because the $7P_{\{3/2\}}$ population is transient during the off-resonant Raman ($\gamma_P T_f \approx 10^{-3}$ at $T_f = 100$ ns); the CCX gate does not couple to $|P\rangle$ at all. A dedicated experimental measurement of τ_P at the Rb $7P_{\{3/2\}}$ configuration the protocol uses would close this gap.
- **Robust score 7/8, not 1.00**. The single failing neighbour in the operating-point anchor's Manhattan-1 ball is $r = 2.8$ (the $r = 3.2$ neighbour now passes under 0 K lifetimes). It is a $\sim 7\%$ off-axis step in the EIT ratio. Real-lab EIT-ratio calibration is typically $\approx 1\%$, so the protocol's effective robustness in deployment is far better than the grid-step number suggests. The 6/8 figure should be read as "the ratio axis is the only one that matters for fine calibration."
- **K-round (Z-sub-cycle) levels still TBD**. The Rb $54D_{\{3/2\}}$ level for the Z-round has not been independently optimized; see `SelfCorrectingRydberg/plan/Rb_Cs_level_redesign.md` § 6.
- **Higher-order leakage at $V_{ct} > \Delta$.** At $a = 4.0$ µm, $V_{ct} = 442$ MHz is comparable to $\Delta = 500$ MHz. The perturbative analytic estimate $K^* - 1 \propto -1/M_2$ used in `Why_K095.typ` is borderline here; the fine-scan numerical result $K^* \approx 1.00$ at $r = 3.0$, $\Delta = 500$ tells us the higher-order correction has flipped sign relative to the $V_{ct} \ll \Delta$ regime that gave $K^* \approx 0.95$.

Bibliography

[1] N. Šibalić, J. D. Pritchard, C. S. Adams, and K. J. Weatherill, "ARC: An open-source library for calculating properties of alkali Rydberg atoms," *Computer Physics Communications*, vol. 220, pp. 319–331, 2017, doi: 10.1016/j.cpc.2017.06.015.

[2] I. I. Beterov, I. I. Ryabtsev, D. B. Tretyakov, and V. M. Entin, "Quasiclassical calculations of blackbody-radiation-induced depopulation rates and effective lifetimes of Rydberg nS, nP, and nD alkali-metal atoms with $nl = 80$", *Physical Review A*, vol. 79, no. 5, p. 52504, 2009, doi: 10.1103/PhysRevA.79.052504.

[3] U. Volz and H. Schmoranzer, "Precision lifetime measurements on alkali atoms," *Physica Scripta*, pp. 48–55, 1996, doi: 10.1088/0031-8949/1996/T65/007.

[4] L. H. Pedersen, N. M. Møller, and K. Mølmer, "Fidelity of quantum operations," *Physics Letters A*, vol. 367, no. 1–2, pp. 47–51, 2007, doi: 10.1016/j.physleta.2007.02.069.

[5] M. A. Nielsen, "A simple formula for the average gate fidelity of a quantum dynamical operation," *Physics Letters A*, vol. 303, no. 4, pp. 249–252, 2002, doi: 10.1016/S0375-9601(02)01272-0.

[6] A. M. Farouk, I. I. Beterov, P. Xu, S. Bergamini, and I. I. Ryabtsev, "Parallel Implementation of CNOT^N and C^2NOT^2 Gates via Homonuclear and Heteronuclear Förster Interactions of Rydberg Atoms", *Photonics*, vol. 10, no. 11, p. 1280, 2023, doi: 10.3390/photonics10111280.